

## Business Mathematics II

BBA — 2nd Semester — 2024 — Answer Sheet

Subject Code: MTH202

**Note:** This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

### Section A

Brief Answer Questions:  $[10 \times 1 = 10]$

**1. Integrate:  $\int \frac{x+2}{x-5} dx$**

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

**2. Find the area of the region bounded by the curve  $y = \frac{3}{2}x^2$ , the  $y$ -axis and the two ordinates  $x = 0$  and  $x = 2$ .**

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

**3. Find  $f_{xx}$  when  $f(x, y) = 3x^2y^2 + 5x^2y$ .**

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

**4. If  $A = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}$ , find  $A^2$ .**

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

**5. Evaluate:  $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{bmatrix}$**

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

**6. The input-output coefficient matrix of an economy of two industries is  $\begin{bmatrix} 0.8 & 0.2 \\ 0.9 & 0.7 \end{bmatrix}$ . Test whether the system is viable as per Hawkins-Simon conditions.**

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

**7. Determine the order and degree of the differential equations:  $\left(\frac{dy}{dx}\right)^2 - \left(\frac{dy}{dx}\right)^5 + 7xy = 0$ .**

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

**8. Solve the differential equation:  $\frac{dy}{dx} = \frac{1}{x}$**

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

**9. Solve the first order homogeneous difference equations:  $y_{t+1} - 2y_t = 0$ ,  $y_0 = 3$**

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

**10. The marginal cost function of manufacturing  $x$  units of a product is given by  $MC = (3x^2 - 10x + 3)$ . The total cost of producing one unit of the product is Rs. 7. Find average cost function.**

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

## Section B

Short Answer Questions: (Attempt any FIVE Questions)  $[5 \times 3 = 15]$

**11. Integrate the following: (a)  $\int \frac{\log x + 5}{x} dx$  (b)  $\int (2x + 1)\sqrt{5x + 2} dx$**

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

**12. The demand function for a product is  $P_d = 8 - 2x$  and the supply function is  $P_s = 2x$  under pure competition. Find the consumer's surplus and producer's surplus.**

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

**13. Solve the following system of linear equations using Cramer's rule or matrix method:  $4x + 6z = 100$ ,  $3x + 6y + z = 100$ ,  $3x + 4y + 3z = 100$**

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

**14. A firm's production function is given by the equation  $Q = 100L^{0.5}K^{0.5}$ . Use partial differentiation to find the approximate change in output  $Q$  when the firm's manager increases capital by 5% and decreases labour by 3%.**

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

**15. Solve the differential equations:  $\frac{dy}{dx} + \frac{1}{x}y = x^3$**

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

**16. The reaction functions for the two duopolists, firms X and Y are given by the equations  $P_t^X = 45 + 0.8P_{t-1}^Y$  and  $P_t^Y = 45 + 0.8P_{t-1}^X$ , respectively. If the assumptions of the Bertrand model hold, derive a difference equation for  $P_t^X$  and calculate what  $P_t^X$  will be in time period 10 if firm X starts off in time period 0 by setting a price of 300.**

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

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17. Find the time path of the national income  $(y_t)$  of the following data and comment on the stability of the time path.  $(y_t = C_t + I_t)$   $(C_t = 200 + 0.75y_{t-1})$   $(I_t = 50 + 0.15y_{t-1})$  and  $(y_0 = 3000)$ .

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

## Section C

Long Answer Questions: (Attempt any THREE Questions)  $[3 \times 5 = 15]$

18. Solve the following LP problem using the simplex method or graphic method: Maximize  $(Z = 5x_1 + 3x_2)$ . Subject to the constraints  $(2x_1 + x_2 \leq 5)$   $(x_1 + x_2 \leq 4)$  and  $(x_1, x_2 \geq 0)$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

19. For the following transaction matrix of two sector economy consisting two industries P and Q, calculate the gross output for each industry if the final demand changes to 18 units for P and 44 units for Q.

| Producer \ User | Final demand | Total output |
|-----------------|--------------|--------------|
| P               | 16           | 20           |
| Q               | 4            | 40           |
| Q               | 8            | 40           |
| P               | 2            | 8            |

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

20. Solve the difference equation:  $(y_t = 0.6y_{t-1} + 21)$   $(y_0 = 50)$ . Find  $(y_5)$  and  $(y_{11})$ ?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

21. In a competitive market price where  $(Q_d = 500 - 5P)$  and  $(Q_s = 40 + 20P)$ , the initial price  $(P(0))$  is Rs 50. (a) Derive a function for the time-path of P and use it to predict price in time period 5 given that price adjusts in proportion to excess demand at the rate  $(\frac{dP}{dt} = 0.01 (Q_d - Q_s))$ . (b) How many time periods would you have to wait for the price to drop by Rs. 30?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

## Section D

Comprehensive Answer / Case / Situation Analysis Questions:  $[2 \times 10 = 20]$

22. The total cost and demand functions for two goods X and Y are given by the equations  $(TC = 10 - 6x + 5y)$   $(P_x = 6 - 2x)$   $(P_y = 25 - 0.1y)$  where x and y are the number of units of the first goods and second goods respectively. a. Write down the equations for total revenue and total profit. b. Determine the number of units of each good which should be sold to maximize revenue. Calculate the maximum revenue and prices of each good under revenue maximizations. c. Determine the number of units of each good which should be bought and sold to maximize profits. d. Calculate the maximum profit, prices of each good and total revenue under profit maximization.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

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These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.